

Git & GitHub

Version control for every project · Equigrow short notes

Why Git

Git saves snapshots (commits) of your project so you can go back, work on features separately (branches) and collaborate. GitHub hosts your repositories online.

Everyday commands

```
git init                # start tracking a folder
git clone <url>          # copy a repository
git status              # what changed?
git add file.txt (or git add .)
git commit -m "Add login page"
git log --oneline       # history
git push               # upload commits
git pull               # download and merge others' commits
```

Branches

```
git branch feature-login # create
git switch feature-login # move to it (older: git checkout)
git switch main
git merge feature-login  # bring the work into main
```

Work on a branch, then open a **pull request** on GitHub so others can review before merging.

Fixing mistakes

Problem	Command
Undo changes to a file (not committed)	git restore file.txt
Unstage a file	git restore --staged file.txt
Fix the last commit message	git commit --amend
Undo a pushed commit safely	git revert <commit-id>

Good habits

- Commit small, related changes with clear messages ("Fix login bug", not "update").
- Add a **.gitignore** so build files and secrets are never committed.
- Write a README: what the project does, screenshots, how to run it. Pin your best repos on your profile.